

PREPRINT DRAFT

Information Asymmetry

Higher-Level Psychological Descriptions as Compressions of Deeper Dynamical Representations

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Central thesis. Psychological descriptions such as anxiety, confidence, motivation, attention, and emotion can be investigated as compressed higher-level summaries of deeper dynamical structure. If the thesis is correct, rich dynamical descriptions should predict psychological descriptions more completely than psychological descriptions can reconstruct the original dynamics.

July 2026

Abstract

Psychological concepts such as anxiety, confidence, motivation, attention, and emotion are widely used to explain and predict human behaviour. This work investigates an alternative representational hypothesis: higher-level psychological descriptions are compressions of deeper dynamical descriptions. If correct, sufficiently rich dynamical representations should consistently generate or predict psychological descriptions, while the reverse reconstruction should lose information. This document presents the hypothesis, formalises the expected information asymmetry, proposes a falsifiable experimental test, introduces an engineering implementation inspired by the same structural principles, and establishes a broader research programme investigating whether common dynamical representations can support prediction, explanation, and early detection across increasingly complex systems.

KEYWORDS

Dynamical systems; psychological description; information asymmetry; state space; reachability; constraints; runtime governance; pre-execution safety; complex systems.

SECTION 1

Motivation

Psychology has developed powerful descriptive concepts for understanding behaviour. Terms such as stress, confidence, motivation, attention, and emotion compress complex patterns into language that humans can communicate, compare, and act upon. These descriptions are not being treated here as mistakes or as useless categories. They are treated as successful high-level summaries.

The question considered here is whether those summaries are the most fundamental representation available for prediction, explanation, and early intervention. A label such as *anxiety* may be clinically, socially, and subjectively meaningful while still failing to preserve the underlying trajectory, constraint structure, feedback dynamics, and reachable futures that produced it. The same label can therefore refer to many distinct underlying dynamical situations.

Research question. Are psychological descriptions fundamental explanations of behaviour, or are they compressed summaries of deeper dynamical structure?

This work explores the second possibility. On this view, a psychological label is retained, but reinterpreted as a higher-level projection of an underlying dynamical representation. The central test is whether the deeper representation contains information that survives forward prediction but is lost under reverse reconstruction. This framing places the proposal in a lineage of information theory, cybernetics, control, systems theory, and multi-level explanation (Shannon, 1948; Wiener, 1948; Ashby, 1956; von Bertalanffy, 1968; Kalman, 1960; Marr, 1982).

Why This Matters

Testing this hypothesis matters because representational choices determine what a theory can explain, what it can predict, and what it can detect early. If psychological labels compress dynamical structure, then

two systems with the same label may still have different reachable futures, different constraints, different intervention points, and different risks. A description that is useful for communication may therefore be insufficient for prevention.

The practical importance is not merely terminological. A richer representation may expose information content that is invisible after compression: which trajectories are becoming unstable, which constraints are tightening, which transitions remain possible, and which future states are becoming reachable. This matters for explanatory power, because the model should account not only for the label assigned to a state but for how that state became reachable. It matters for prediction, because future behaviour often depends on structure that a label does not preserve. It matters for early detection, because harmful outcomes may be preceded by measurable changes in trajectory geometry before they become visible as categorical labels.

A successful programme would not replace every discipline with one vocabulary. Rather, it would test whether common dynamical representations can serve as a translation layer across disciplines: preserving enough structure to compare engineered systems, biological processes, ecological dynamics, financial risk, and human behaviour without collapsing them into a single metaphor. The aim is modest but consequential: determine whether a shared structural representation can carry more predictive and explanatory information than the higher-level descriptions derived from it.

SECTION 2

Core Hypothesis

Let $\mathcal{D}$ denote a dynamical representation and $\mathcal{P}$ denote a psychological description. A dynamical representation may include state space, trajectories, constraints, feedback, temporal ordering, transition structure, and reachability. A psychological description may include labels such as anxiety, confidence, motivation, attention, or emotional state.

$$\mathcal{D} = (\mathcal{X}, x_0, \mathcal{T}, C, F, \mathcal{R}, \Omega), \quad (1)$$

$$\mathcal{P} = \{p_1, p_2, \dots, p_k\}. \quad (2)$$

The hypothesis is:

$$\textbf{Higher-level psychological descriptions are compressions of deeper dynamical descriptions.} \quad (3)$$

If the hypothesis is correct, then two directional properties should hold:

1. A forward mapping from dynamics to psychology should be learnable, stable, and predictive.
2. A reverse mapping from psychology to dynamics should be lossy, underdetermined, and structurally incomplete.

Formally, if

$$f : \mathcal{D} \rightarrow \mathcal{P} \quad (4)$$

and

$$g : \mathcal{P} \rightarrow \widehat{\mathcal{D}}, \quad (5)$$

then the claim predicts that, for relevant complex systems,

$$d(\mathcal{D}, \widehat{\mathcal{D}}) > \varepsilon \quad \text{and} \quad H(\mathcal{D} | \mathcal{P}) > 0, \quad (6)$$

where d measures structural reconstruction error, ε is a pre-registered threshold, and $H(\mathcal{D} | \mathcal{P})$ denotes residual uncertainty about the original dynamical representation after only the psychological description is retained.

SECTION 3

Information Asymmetry

The expected asymmetry is directional. The dynamical representation should contain enough structure to predict the psychological description. The psychological description should not contain enough information to reconstruct the full original dynamics.

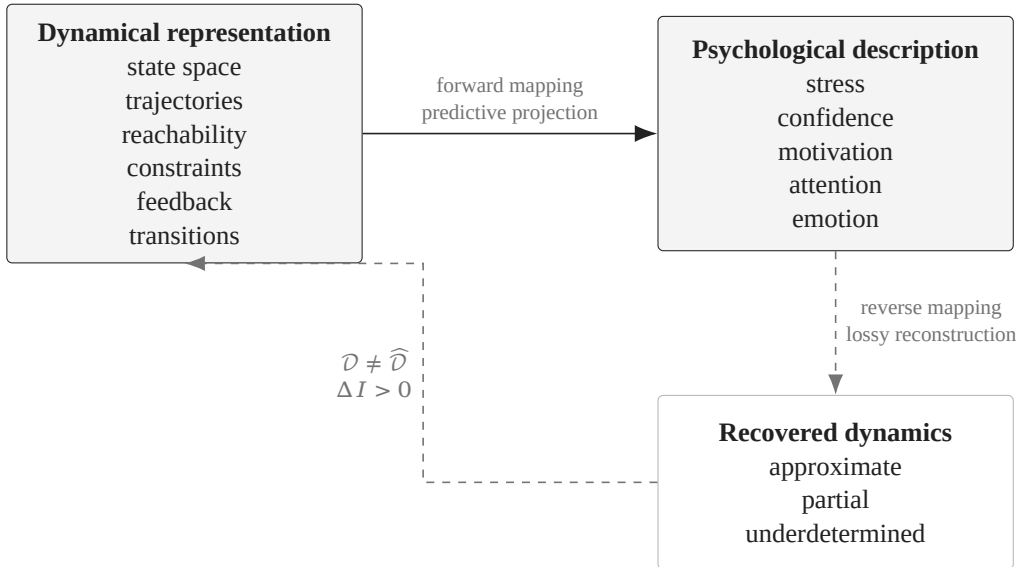


Figure 1: The proposed information asymmetry. Dynamics can project into psychological language, while psychological language is expected to reconstruct only an approximate subset of the original dynamics.

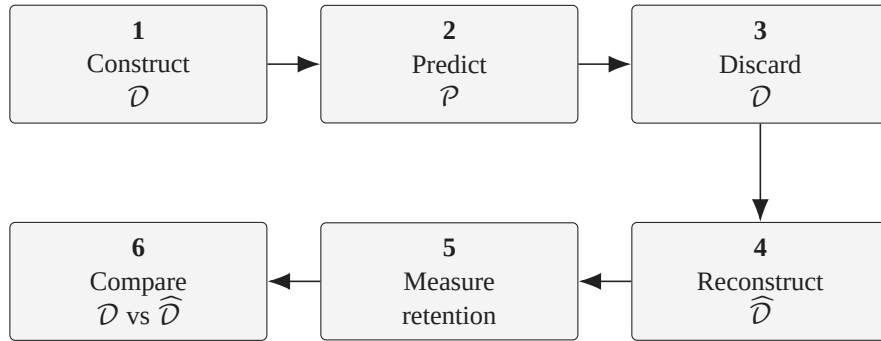
The compression claim becomes meaningful only when it is tested against recoverable structure. If the psychological description preserves the transition topology, temporal ordering, constraint geometry, and

reachability properties of the original dynamics, the compression claim weakens. If it cannot preserve those properties, the original representation contains explanatory information not carried by the higher-level label.

SECTION 4

Proposed Experimental Test

The proposed test is deliberately simple. It does not begin with psychological labels. It begins with the dynamical representation alone. The experiment asks whether a higher-level description can be generated from the structure, then asks whether the structure can be recovered after the original representation has been removed.



A positive result requires strong forward prediction and measurable reverse-reconstruction loss under pre-registered thresholds.

Figure 2: *Experimental protocol. The asymmetry is measured by predicting psychological descriptions from dynamics, retaining only the descriptions, and then testing how much of the original dynamics can be recovered.*

Protocol

1. **Construct the dynamical representation.** Build $\mathcal{D}$ from state variables, trajectories, constraints, transitions, feedback relations, and reachability properties.
2. **Predict psychological descriptions.** Estimate $\mathcal{P}$ from $\mathcal{D}$ without using the original labels as primitive explanatory variables.
3. **Discard the dynamics.** Retain only the psychological description $\mathcal{P}$.
4. **Attempt reverse reconstruction.** Use $\mathcal{P}$ to reconstruct $\widehat{\mathcal{D}}$.
5. **Measure retained information.** Compare $\mathcal{D}$ and $\widehat{\mathcal{D}}$ using structural, temporal, and predictive metrics.

Falsification criterion

The hypothesis is weakened if psychological descriptions alone can reconstruct the relevant dynamical properties at or above the pre-registered threshold across multiple tasks, subjects, and contexts. The hypothesis is strengthened if the forward mapping is predictive while the reverse mapping consistently fails to recover

structural information. The important point is that the proposal is not defended by rhetorical preference for one vocabulary over another. It is defended only if a measurable representational asymmetry appears.

SECTION 5

Expected Information Loss

The reverse mapping is expected to lose structural detail because psychological descriptions compress many possible dynamical states into a smaller semantic space. Multiple distinct trajectories may all be described as *anxiety*; multiple distinct constraint configurations may all be described as *low confidence*; multiple distinct feedback patterns may all be described as *motivation loss*. The label may be accurate while still being structurally incomplete.

Dynamical feature	Expected reverse-mapping failure
Reachability	Which future states were reachable, blocked, or conditionally reachable may not be recoverable from the label alone.
Constraint structure	The boundaries shaping behaviour may be compressed into a single psychological term.
Trajectory geometry	The path by which the state moved through time may be lost.
Transition topology	The permitted and forbidden transitions between states may be underdetermined.
Temporal ordering	The sequence of changes preceding the label may not be recoverable.
Feedback dynamics	Reinforcing and stabilising loops may be collapsed into one description.
Degrees of freedom	The number of independent controllable variables may be hidden by the label.

Table 1: Representative information expected to be lost when dynamics are compressed into psychological descriptions.

The resulting information loss can be expressed as:

$$\Delta I = I(\mathcal{D}; \mathcal{D}) - I(\mathcal{D}; \widehat{\mathcal{D}}), \quad (7)$$

where a positive ΔI indicates that the recovered dynamics preserve less information about the original dynamics than the original representation contained. Empirically, this loss should be measured through structural reconstruction error, predictive degradation, missing transitions, unrecovered constraints, and failure to reproduce the reachable set.

Engineering Inspiration

The same structural principle motivated an engineering implementation based on dynamics rather than psychological concepts. Instead of reasoning about emotions, intentions, or narrative explanations, the implementation evaluates trajectories, constraints, reachability, forbidden regions, allowable transitions, and execution geometry before actions occur.

Engineering translation. If catastrophic outcomes are preceded by unsafe structural trajectories, then useful governance can occur before execution by identifying whether an action path enters a forbidden region of state space.

This translation does not claim that autonomous-system safety and human psychology are the same object. It claims something narrower: a structural representation can be useful when a domain has states, actions, transitions, constraints, and forbidden outcomes. Autonomous systems are an appropriate first engineering domain because these elements can be specified, evaluated, and audited more directly than in many biological or psychological settings.

The implementation therefore serves as a practical testbed for related structural principles. It does not prove the representational hypothesis about psychology. It shows that the vocabulary of trajectories, constraints, reachability, and forbidden regions can produce operational behaviour in a measurable engineering environment.

Runtime Governance

The engineering implementation became Runtime Governance. Its central principle is:

Prevent unsafe trajectories before execution. (8)

Instead of classifying behaviour psychologically, the system governs execution paths, state transitions, reachable futures, and constraint violations using structural representations.

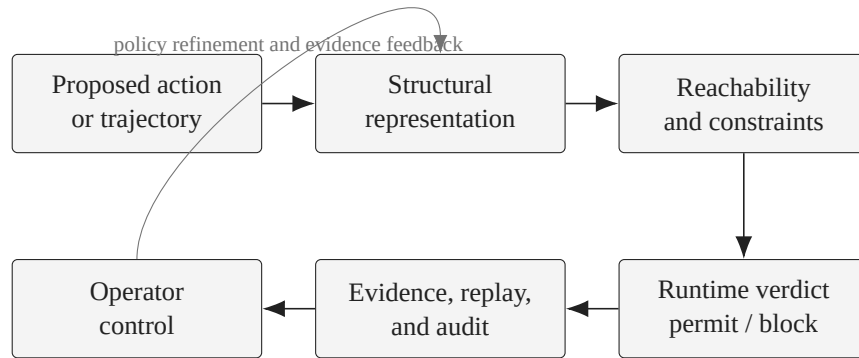


Figure 3: Runtime Governance applies the structural view to autonomous systems: proposed actions are converted into trajectory representations, evaluated against constraints and forbidden regions, and governed before execution.

The governing condition can be expressed as:

$$\text{Safe}(E, t) \iff \mathcal{R}_E(t) \cap \Omega = \emptyset, \quad (9)$$

where E is an environment or execution context, $\mathcal{R}_E(t)$ is the reachable set at time t , and Ω is the set of forbidden or catastrophic states. In this setting, the central representational question becomes operational: does the proposed action path enter a region that should not be reachable? The answer can be evaluated before execution, logged as evidence, and replayed for audit.

Live Engineering Demonstration

A public implementation of Runtime Governance is available through the Resurrection Tech website and accompanying Live Demo, where readers can explore the engineering system and interact with a live trajectory evaluation demonstrating pre-execution governance in practice. The demonstration is an engineering artefact, not proof of the psychological compression hypothesis; it shows that an implementation inspired by the same structural framework is operational, publicly observable, and testable in a measurable autonomous-system setting.

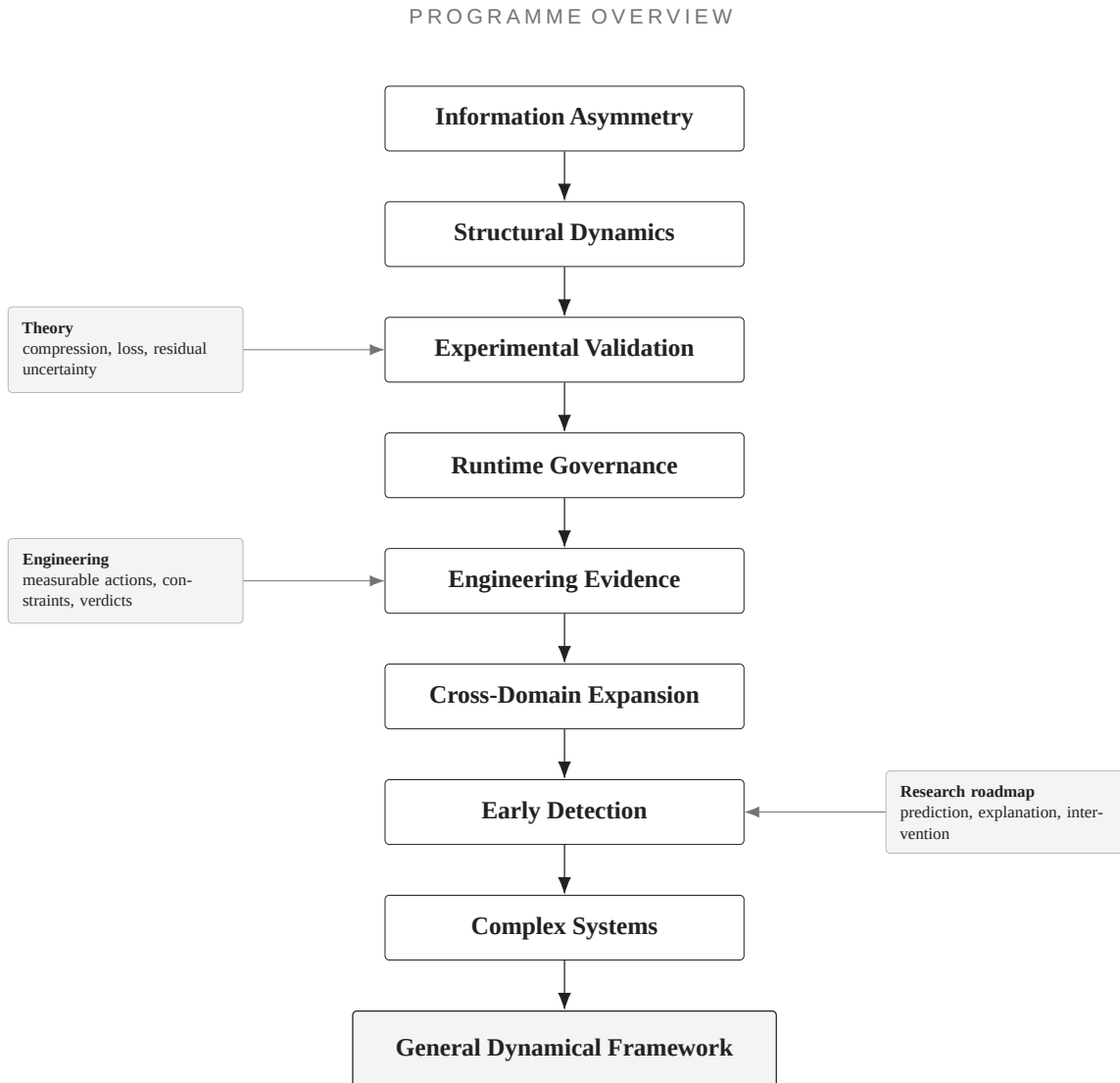


Figure 4: Programme overview. The paper begins with a representational asymmetry, formalises it through structural dynamics, proposes experimental validation, and then connects the same structural principles to Runtime Governance, engineering evidence, and cross-domain expansion toward early detection in complex systems.

Engineering Evidence

The implementation demonstrates the practical utility of structural representations for governing autonomous systems. The evidence is engineering evidence: live deployment, deterministic evaluation, pre-execution trajectory governance, production onboarding, organisational runtime control, multi-domain policy enforcement, and repeatable behaviour.

Observed capability	Structural significance
Live production deployment	The framework can operate outside a static paper model.
Deterministic runtime evaluation	Governance behaviour can be replayed and audited.
Pre-execution trajectory governance	Unsafe paths can be blocked before actions reach production systems.
Production onboarding	New organisations and environments can be provisioned into the governance layer.
Organisational runtime control	Operators can move from assessment to enforcement and evidence generation.
Multi-domain enforcement	The same structural approach can be applied across different risk domains.
Repeatable engineering behaviour	The representation supports consistent evaluation rather than ad hoc interpretation.

Table 2: *Engineering evidence supporting the utility of structural representations in autonomous-system governance.*

The engineering system does not prove the psychological hypothesis. It demonstrates that the structural framework is operationally useful in one measurable domain: autonomous-system safety. That domain becomes the first testbed for broader representational claims, because the relevant entities can be made explicit: action proposals, state transitions, policies, forbidden regions, operator decisions, and runtime verdicts.

Relationship Between Theory and Engineering

The relationship between the theory and the engineering implementation is sequential rather than identical. The hypothesis concerns representational compression. The engineering implementation applies related structural principles to a domain where trajectories, constraints, and forbidden states can be measured directly.

This matters for the research programme because it prevents an overclaim. Runtime Governance is not presented as proof that psychological descriptions are compressions of dynamics. It is presented as the first engineering implementation of related structural principles. The implementation demonstrates that structural

representations can support prediction, constraint evaluation, and early blocking in a concrete domain. That result motivates, but does not replace, the controlled experiments required to test the representational hypothesis.

Bridge claim. The hypothesis supplies the representational question; Runtime Governance supplies an engineering testbed where related structural primitives can be made operational and measured before execution.

The research sequence is therefore: formalise the asymmetry, design the experiment, validate the forward and reverse mappings, apply the structural vocabulary where it is measurable, and then test whether the same representational advantages generalise across domains.

SECTION 10

Future Research Programme

The long-term objective extends beyond autonomous systems. Autonomous systems provide the first implementation because they allow structural representations to be tested in measurable engineering environments. Proposed actions can be represented as trajectories. Policies can define constraints. Forbidden regions can be specified. Runtime verdicts can be logged. Failures can be replayed. This makes autonomous systems an unusually practical first domain for studying structural prediction and pre-execution prevention.

The broader research programme asks whether the same representational principles can support prediction, explanation, and early detection across increasingly complex systems. The programme should proceed carefully, moving from domains with explicit state and action representations toward domains where the state space is harder to observe, noisier, and more ethically sensitive.

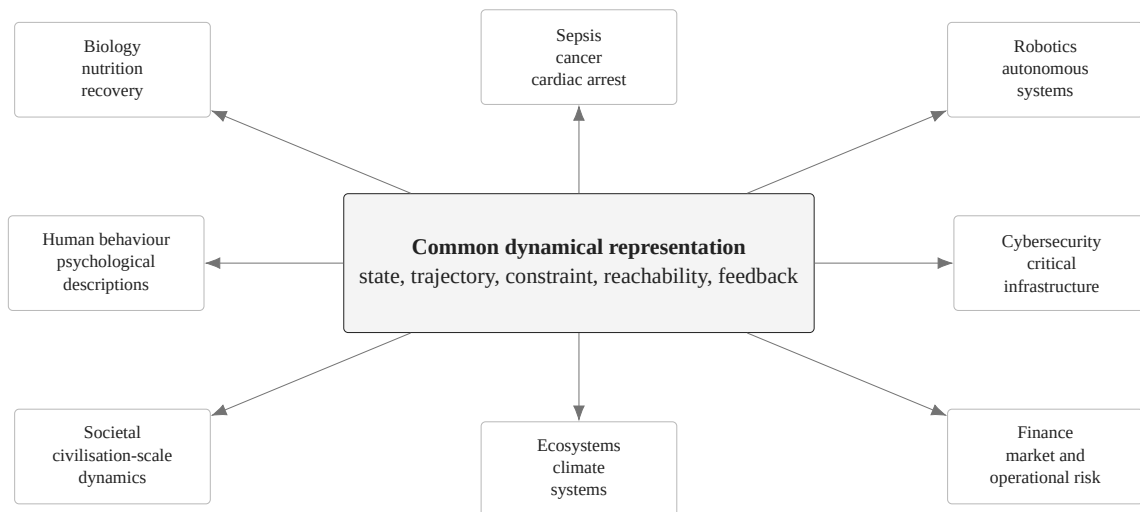


Figure 5: Cross-domain roadmap. Autonomous systems are the first implementation because they provide measurable trajectories, constraints, and verdicts. Later domains require stronger validation, richer data, and domain-specific safeguards.

Initial engineering domains

The first class of domains should be those where actions, policies, and failure states can be defined with relative precision. These include robotics, autonomous systems, cybersecurity, finance, and critical infrastructure. In these domains, the relevant question is whether structural representations can identify risky trajectories earlier than label-based monitoring alone.

Biological and medical domains

The next class of domains includes biology, nutrition, sepsis, cancer, cardiac arrest, ACL injury prediction, recovery modelling, and personalised medicine. These domains are attractive because they contain trajectories, constraints, feedback loops, and early-warning signals. They are also more difficult because the relevant state variables may be partially observed, noisy, ethically sensitive, and heterogeneous across individuals. Any extension into these areas would require careful empirical validation, clinical collaboration, and strict separation between hypothesis generation and medical decision-making.

Ecological and societal domains

A third class includes ecosystems, climate systems, and civilisation-scale dynamics. These domains test the limits of the framework. They involve many interacting agents, delayed feedback, nonlinear transitions, unknown thresholds, and partial observability. Here the goal would not be deterministic prediction of every outcome. It would be to determine whether structural representations improve detection of dangerous trajectories, constraint collapse, instability, and transitions toward undesirable reachable regions.

Research roadmap

Phase	Primary question	Representative domains
I. Formalisation	Can the asymmetry be defined and measured?	Psychological labels, simulated systems, controlled behavioural tasks.
II. Engineering validation	Can structural representations govern measurable trajectories before execution?	Autonomous systems, robotics, cybersecurity, finance.
III. Biological extension	Can dynamical structure improve early detection under noisy observation?	Nutrition, sepsis, cancer, cardiac arrest, injury prediction, recovery modelling.
IV. Infrastructure and ecology	Can the framework identify unstable trajectories in large coupled systems?	Critical infrastructure, ecosystems, climate systems.
V. General framework	Can a common representation support cross-domain prediction and explanation?	Complex engineered, biological, ecological, and societal systems.

Table 3: *A staged roadmap for testing whether common dynamical representations can support prediction, explanation, and early detection across progressively harder domains.*

The programme should remain falsifiable at each stage. A domain should be added only when its state variables, transition rules, constraints, relevant labels, and evaluation metrics can be specified clearly enough to support measurement. The aim is not to force a universal language onto every system. The aim is to determine where a common dynamical representation genuinely preserves useful information that higher-level labels lose.

SECTION 11

Long-Term Vision

The objective is to investigate whether a common dynamical representation can provide a more fundamental explanatory framework across complex systems. Rather than waiting for catastrophic outcomes, the framework seeks to identify the structural trajectories that precede them.

The long-term ambition is not merely better classification after a state has already become visible. It is earlier detection of the pathways by which harmful states become reachable. In an engineered system, that may mean blocking a proposed action before it reaches production. In a biological system, it may mean identifying a trajectory of deterioration before a categorical diagnosis becomes obvious. In an ecological or societal system, it may mean recognising constraint collapse, feedback amplification, or transition risk before a crisis is fully expressed.

This paper should therefore be read as the opening of a research programme rather than the closure of an argument. The first claim is representational: some higher-level descriptions may be compressed projections

of deeper dynamics. The first test is experimental: determine whether forward prediction from dynamics succeeds while reverse reconstruction from labels loses measurable information. The first implementation is engineering: use structural representations to govern autonomous-system trajectories before execution. The future work is cross-domain: test whether similar structural advantages appear in increasingly complex systems.

If successful, the programme would offer a disciplined way to move from description toward prevention. It would preserve the value of high-level language while asking where that language loses information, where earlier structure remains visible, and where intervention becomes possible before collapse becomes irreversible.

SECTION 12

Conclusion

The central question is simple:

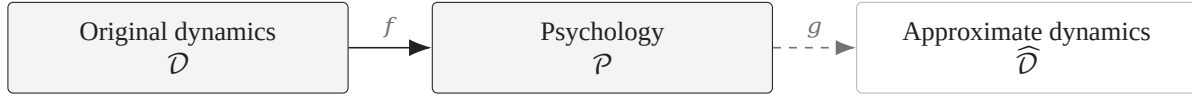
Can a dynamical description reproduce the explanatory and predictive role of psychological concepts across diverse situations while also making accurate predictions that psychological labels alone cannot?

If the answer is yes, psychological descriptions become interpretable as compressed representations of deeper dynamical structure. This document proposes a framework for investigating that possibility through theory, experiment, engineering implementation, and future research.

The immediate test is not rhetorical. It is representational. Start with dynamics. Predict psychological descriptions. Remove the dynamics. Attempt reconstruction. Measure what is lost. If the recovered dynamics are incomplete, the asymmetry is visible.

This paper introduces a testable representational hypothesis, connects it to an engineering implementation built from related structural principles, and opens a research programme on whether common dynamical representations can improve prediction, explanation, and early detection across engineered, biological, ecological, and societal systems. Its central promise is not that one framework can explain everything, but that better representations may reveal dangerous trajectories earlier - before labels harden, before outcomes arrive, and before preventable collapse becomes irreversible.

Appendix: Compact Statement of the Test



$$\mathcal{D} \neq \widehat{\mathcal{D}} \quad \text{and} \quad \Delta I > 0$$

Figure 6: Compact form: dynamics project into psychology; psychology reconstructs only approximate dynamics; positive information loss supports the compression hypothesis.

Minimum empirical commitments

A strong test should pre-register: the dynamical variables, the psychological labels, the forward-prediction metric, the reverse-reconstruction metric, the threshold ε , the domains tested, and the conditions under which the hypothesis is weakened or rejected.

Recommended reconstruction metrics

Potential metrics include structural edit distance over transition graphs, divergence between reachable sets, predictive loss under held-out trajectories, missing-constraint count, temporal-ordering error, and degradation in intervention selection. No single metric is sufficient. A mature experiment should report several, because a psychological description might preserve one form of information while losing another.

Interpretive caution

A positive information asymmetry would not show that psychological language is invalid. It would show that the language is compressed relative to the selected dynamical representation and task. Psychological descriptions may remain useful for communication, lived experience, clinical interpretation, social meaning, and decision-making. The claim concerns information content and reconstruction, not human worth, experience, or dignity.

Selected Foundational References

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